

Exercise 1- Mocking and Stubbing

ExternalApi.java

```
package com.example;

public interface Externalapi {

    String getData();

}
```

MyService.java

```
package com.example;

public class myservice {

    private Externalapi api;

    public myservice(Externalapi api) {

        this.api = api;

    }

    public String fetchData() {

        return api.getData();

    }

}
```

MyServiceTest.java

```
package com.example;

import org.junit.Test;
```

```

import static org.junit.Assert.assertEquals;

import static org.mockito.Mockito.*;

public class myservicetest {

    @Test

    public void testExternalApi() {

        Externalapi mockApi = mock(Externalapi.class);

        when(mockApi.getData()).thenReturn("Mock Data");

        myservice service = new myservice(mockApi);

        String result = service.fetchData();

        assertEquals("Mock Data", result);

    }

}

```

The screenshot shows an IDE with two tabs: 'myservice.java' and 'myservicetest.java'. The 'myservicetest.java' tab is active, displaying the same Java code as the previous block. Below the code editor, the 'TERMINAL' tab is open, showing the output of a test run. The output indicates that the tests passed successfully.

```

[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.345 s
[INFO] Finished at: 2026-06-25T15:07:53+05:30
[INFO] -----

```

The status bar at the bottom of the IDE shows 'Ln 28, Col 2', 'Spaces: 4', 'UTF-8', 'CRLF', and 'Java'.

Exercise 2-Verifying Interactions

VerifyInteractionTest.java

```
package com.example;

import org.junit.Test;

import static org.mockito.Mockito.*;

public class VerifyInteractionTest {

    @Test

    public void testVerifyInteraction() {

        Externalapi mockApi = mock(Externalapi.class);

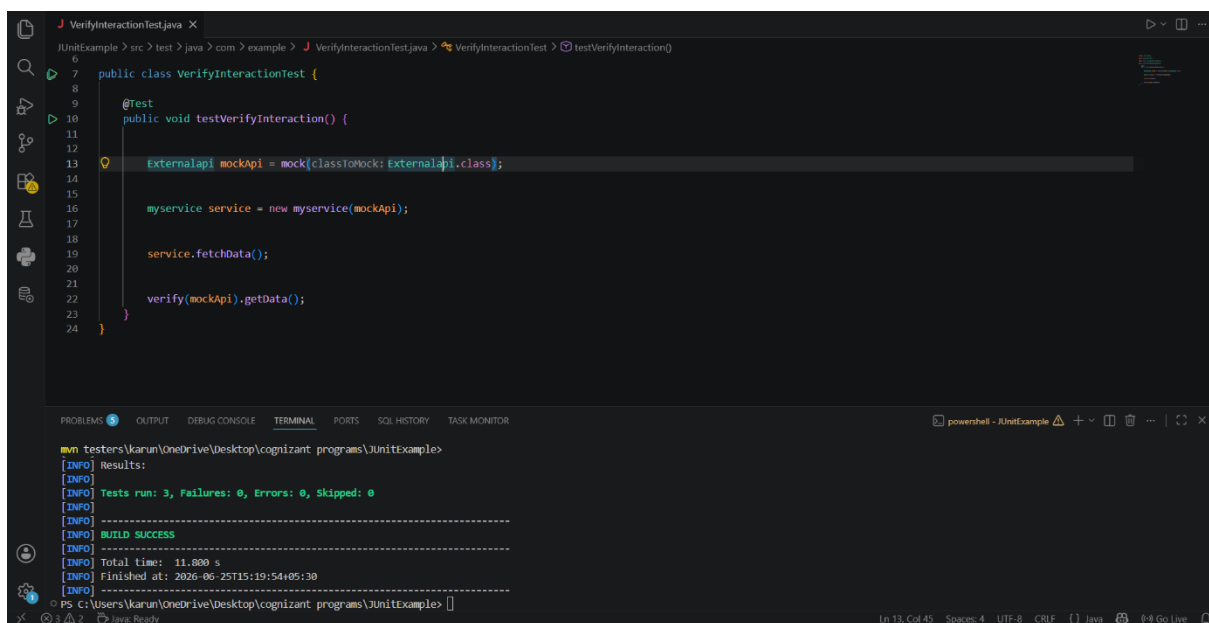
        myservice service = new myservice(mockApi);

        service.fetchData();

        verify(mockApi).getData();

    }

}
```



The screenshot shows an IDE window with the file `VerifyInteractionTest.java` open. The code is as follows:

```
public class VerifyInteractionTest {
    @Test
    public void testVerifyInteraction() {
        Externalapi mockApi = mock(classToMock: Externalapi.class);

        myservice service = new myservice(mockApi);

        service.fetchData();

        verify(mockApi).getData();
    }
}
```

The terminal output shows the results of running the tests:

```
mvn testers\karun\OneDrive\Desktop\cognizant programs\JUnitexample>
[INFO] Results:
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 11.800 s
[INFO] Finished at: 2020-06-25T15:19:54+05:30
[INFO]
```